## SUPPLEMENTARY MATERIAL

**Supplementary Table 1.** Operationalization of modifiable risk factors in the Dutch Brain Research Registry.

| Risk factor | Dutch Brain Research Registry (online self-report) |
|---|---|
| 1. Low cognitive activity | Single question asking whether the participant in general engages in mentally stimulating activities (e.g., learning new things, making music, or pursuing hobbies). Low cognitive activity is defined as a negative response. |
| 2. High cholesterol | History of high cholesterol. |
| 3. Depression | Current or past diagnosis of depression and/or presence of depressive symptoms, based on (1) a mood statement asking whether the participant generally feels cheerful and finds life worthwhile, and (2) a checklist asking which psychiatric or psychological disorders they have or have had, including "depression". |
| 4. Physical inactivity | Single question asking whether the participant engages in physical activity at least twice per week that makes them (slightly) sweat or become short of breath. Physical inactivity is defined as answering less than "two times per week or more". |
| 5. Smoking | Data on current smoking, with response options "no, never", "no, in the past", "yes, currently". The last option was assigned to smoking. |
| 6. Diabetes | History of diabetes. |
| 7. Hypertension | History of hypertension and last measured blood pressure. Participants indicate whether they currently have or have ever had "hypertension (high blood pressure)" and report their last measured systolic (upper) and diastolic (lower) blood pressure values. |
| 8. Obesity | BMI >=30 $kg/m^2$, calculated from self-reported height and weight obtained via questions on current height and current weight. |
| 9. Excessive alcohol use | Participants are asked whether they ever drink alcohol, how often they drink, and how many glasses they usually drink when they do. From this, weekly intake is calculated; high alcohol consumption is categorized as >7 glasses of alcohol per week. |
| 10. Unhealthy diet | Single question asking whether the participant follows a Mediterranean-style diet (e.g., regularly consuming fish, vegetables, legumes, olive oil, and pasta, and occasionally drinking red wine). Unhealthy diet is defined as a negative response. |

*Abbreviations: BMI = Body Mass Index.*

**Supplementary Table 2.** PRS construction parameters based on the largest and most recent (multi-ancestry) GWASs available for each trait.

| Modifiable risk factor | Polygenic risk score (PRS) | First author, year | Total SNPs | SNPs in PRS |
|---|---|---|---|---|
| 1. Low cognitive activity | Educational attainment (in years)$^{\infty}$ | Okbay, 2022[1] | 51,104 | 812 |
| 2. High cholesterol | LDL cholesterol | Graham, 2021[2] | 112,583 | 927 |
| 3. Depression | Major depressive disorder | McIntosh, 2024[3] | 13,074 | 322 |
| 4. Physical inactivity | Days per week of vigorous activity$^{\infty}$ | Karczewski, 2025[4] | 1,244 | 17 |
| 5. Smoking | Cigarettes per day | Saunders, 2022[5] | 4,687 | 70 |
| 6. Diabetes | Type 2 diabetes | Suzuki, 2024[6] | unknown | 1,289 |
| 7. Hypertension | Systolic blood pressure | Keaton, 2024[7] | 94,521 | 1,165 |
| 8. Obesity | Body mass index (BMI) | Kurki, 2023[8] | 7,798 | 114 |
| 9. Excessive alcohol use | Alcohol drinks per week | Saunders, 2022[5] | 11,733 | 140 |
| 10. Unhealthy diet | PC1*/ healthy dietary pattern$^{\infty}$ | Cole, 2020[9] | 9,831 | 151 |

PRSs were computed from GWAS summary statistics based on the largest and most recent (multi-ancestry) GWASs available for each trait. Unless otherwise stated, variants were selected by genome-wide significance ($P \leq 5\times10^{-8}$) followed by LD clumping (±250 kb) using an LD threshold of $r^2 < 0.01$. Only variants passing the *P*-value threshold were retained. Summary-statistic variant extraction and harmonization were performed using the pipeline described in the corresponding reference. Study-specific PRS construction: *Suzuki et al. (2024)*: independent index variants defined at $P \leq 5\times10^{-8}$ using LD clumping within ±5 Mb and an LD threshold of $r^2 < 0.05$. *PC1 is the first principal-component dietary pattern derived from food-frequency items (a "healthy vs. less-healthy" axis primarily defined by whole meal vs. white bread, plus fruits/veg, oily fish, processed meat, etc. ∞ PRS is reversed according to the risk factor. *Abbreviations: APOE = Apolipoprotein E; GWAS = genome-wide association analysis; LD = linkage disequilibrium; PRS = polygenic risk score; SNP = single nucleotide polymorphism*.

**References**


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2. Graham SE, Clarke SL, Wu KHH, et al. The power of genetic diversity in genome-wide association studies of lipids. *Nature*. 2021;600(7890):675-679. doi:10.1038/s41586-021-04064-3
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**Supplementary Table 3.** Numbers of identification of individuals with prevention potential.

| | | Positive online pre-screening | | | | Negative online pre-screening | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | High/intermediate PRS | | Low PRS | | High PRS | | Low/intermediate PRS | |
| **Modifiable risk factor** | **Total *N*** | ***N*** | **%** | ***N*** | **%** | ***N*** | **%** | ***N*** | **%** |
| 1. Low cognitive activity | 953 | 91 | 9.5 | 15 | 1.6 | 165 | 17.3 | 682 | 71.6 |
| 2. High cholesterol | 990 | 228 | 23.0 | 30 | 3.0 | 117 | 11.8 | 615 | 62.1 |
| 3. Depression | 963 | 120 | 12.5 | 21 | 2.2 | 153 | 15.9 | 669 | 69.5 |
| 4. Physical inactivity | 954 | 181 | 19.0 | 54 | 5.7 | 151 | 15.8 | 568 | 59.5 |
| 5. Smoking | 990 | 21 | 2.1 | 6 | 0.6 | 193 | 19.5 | 770 | 77.8 |
| 6. Diabetes | 990 | 60 | 6.1 | 5 | 0.5 | 165 | 16.7 | 760 | 76.8 |
| 7. Hypertension | 990 | 267 | 27.0 | 42 | 4.2 | 116 | 11.7 | 565 | 57.1 |
| 8. Obesity | 988 | 123 | 12.4 | 17 | 1.7 | 164 | 16.6 | 684 | 69.2 |
| 9. Excessive alcohol use | 988 | 214 | 21.7 | 48 | 4.9 | 143 | 14.5 | 583 | 59.0 |
| 10. Unhealthy diet | 952 | 231 | 24.3 | 49 | 5.1 | 143 | 15.0 | 529 | 55.6 |

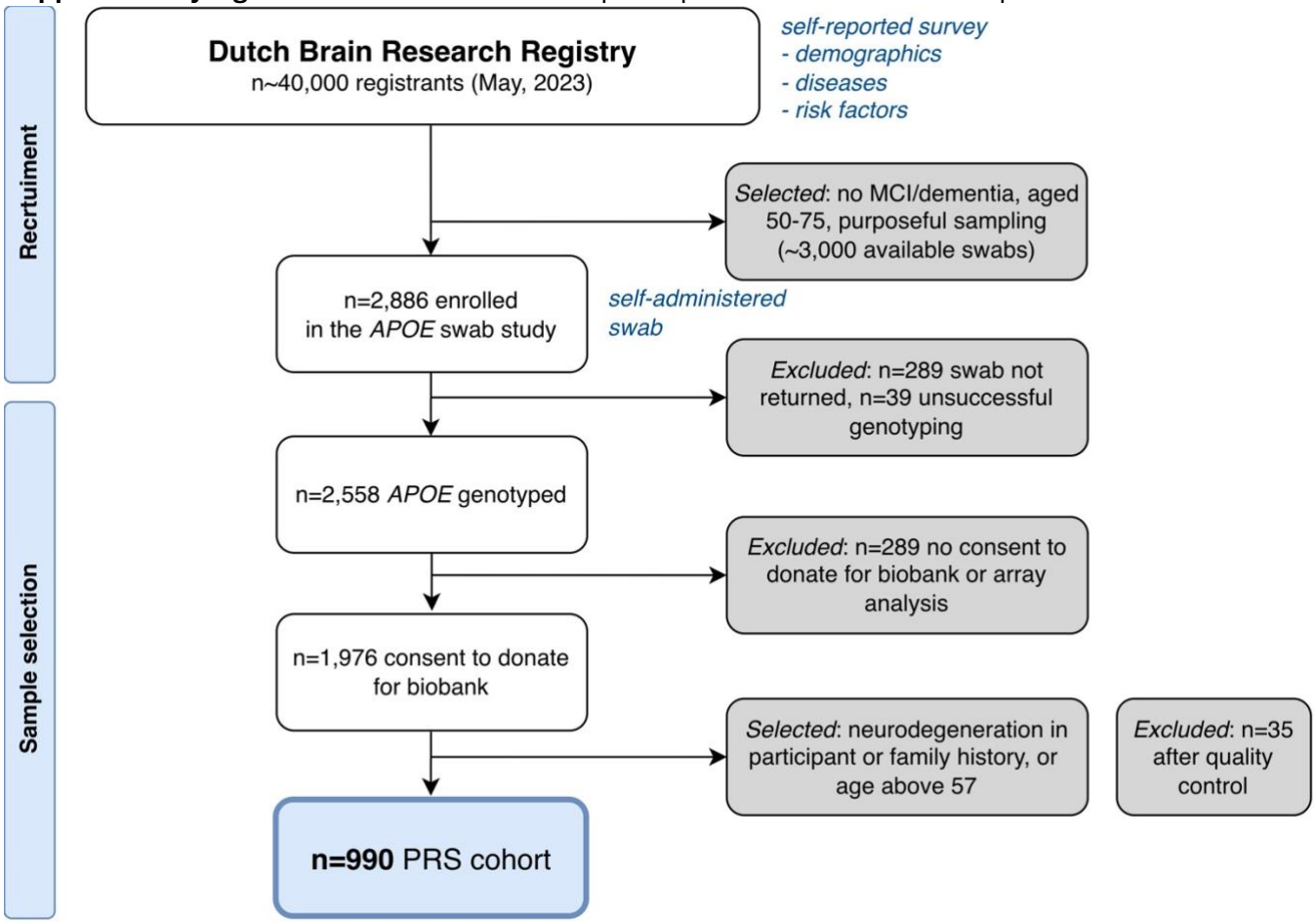
Individuals were assigned to one of four groups: online pre-screening positive with high or intermediate PRS risk; online pre-screening positive with low PRS risk; online pre-screening negative with high PRS risk; and online pre-screening negative with intermediate or low PRS risk.

**Supplementary Table 5.** GWAS of traits underlying polygenic risk scores (PRS) and association with online pre-screening status.

| Modifiable risk factor | GWAS trait used to derive PRS | Variants in **PRS** (SNPs) | Association with online pre-screening | | | |
|---|---|---|---|---|---|---|
| | | | **Continuous PRS** per 1 SD increase | | **Extreme PRS group** highest vs lowest quintile | |
| | | | **OR [95% CI]** | ***P*** | **OR [95% CI]** | ***P*** |
| 1. Low cognitive activity | Educational attainment∞ | 812 | 1.22 [0.99 - 1.50] | 0.0645 | 1.92 [0.94 - 3.95] | 0.0742 |
| 2. High cholesterol | LDL cholesterol | 927 | 1.52 [1.31 - 1.78] | <0.0001 | 3.74 [2.26 - 6.18] | <0.0001 |
| 3. Depression | Major depressive disorder | 322 | 1.25 [1.03 - 1.51] | 0.0238 | 1.91 [1.02 - 3.57] | 0.0418 |
| 4. Physical inactivity | Days per week of vigorous activity∞ | 17 | 0.84 [0.72 - 0.98] | 0.0302 | 0.67 [0.39 - 1.16] | 0.1552 |
| 5. Smoking | Cigarettes per day | 70 | 0.99 [0.67 - 1.47] | 0.9634 | 0.82 [0.24 - 2.78] | 0.7465 |
| 6. Diabetes | Type 2 diabetes | 1,289 | 2.13 [1.58 - 2.86] | <0.0001 | 7.94 [2.93 - 21.53] | <0.0001 |
| 7. Hypertension | Systolic blood pressure | 1,165 | 1.48 [1.28 - 1.72] | <0.0001 | 3.05 [1.90 - 4.90] | <0.0001 |
| 8. Obesity | Body mass index (BMI) | 114 | 1.24 [1.03 - 1.5] | 0.0223 | 2.08 [1.08 - 3.98] | 0.0278 |
| 9. Excessive alcohol use | Alcohol drinks per week | 140 | 1.06 [0.92 - 1.23] | 0.4115 | 1.17 [0.74 - 1.86] | 0.4992 |
| 10. Unhealthy diet | Healthy diet∞ | 151 | 0.99 [0.86 - 1.15] | 0.9073 | 0.99 [0.61 - 1.59] | 0.9562 |

Polygenic risk scores (PRS) were constructed using genome-wide association study (GWAS) summary statistics for each modifiable risk factor. Associations with online pre-screening were tested using logistic regression in the PRS cohort (n = 990), adjusted for age, sex, and the first three genetic PCs. PRSs were analyzed as a standardized continuous predictor (per 1 SD increase) and as an extreme-group contrast (highest vs lowest quintile of the PRS distribution). *Abbreviations: APOE = apolipoprotein E; CI = confidence interval; GWAS = genome-wide association study; NA = not available; OR = odd's ratio; SD = standard deviation; SNP = single nucleotide polymorphism. ∞PRS is reversed according to the risk factor.*

**Supplementary Figure 1.** Extended flowchart of participant recruitment and sample selection in the PRS cohort.




In the *APOE* swab study, 2,886 registrants (aged 50-75 years) had been enrolled to provide a self-administered buccal swab; 2,558 returned a valid sample and were successfully genotyped. Of these, 1,976 provided consent for biobanking and genome-wide single nucleotide polymorphism (SNP) array analysis. SNP array genotyping was prioritized due to funding constraints for registrants who reported evidence of neurodegeneration in themselves or a first-degree relative, or who were aged ≥57 years. In total, 990 individuals underwent SNP array genotyping. *Abbreviations: APOE = apolipoprotein E; DBRR = Dutch Brain Research Registry; MCI = mild cognitive impairment; PRS = polygenic risk score.*